MATH 20250 HOMEWORK 3 NOTES AND HINTS

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1. About Writing

A mathematical response with only numbers and formulas is confusing and hardly-readable. There should at least be descriptions of the formulas suggesting which deduces which, what role the formula X has in the whole proof, etc. Adding such description, like "consequently, we have ...", "this system of equations solve into ...", "performing elementary row operations, we get ..." helps improve the readability of a response, and can never hurt a response.

> Problems are meant to help learn how to write. —- Prof. Kazuya Kato

Any aspiring student in mathematics thus should try to be both rigorous and readable in their written responses to problems. More generally, the skill of producing readable mathematical writings is going to be useful for anyone who is to have a career in quantitative science.

2. Q2.5.5 Hint

Write $u+v+w$, $v+w$, and w as column vectors, in relation to the basis u, v, w. Then, we need only to prove that the resulting matrix in its row echelon form has a pivot in every row, and then apply Proposition 2.3.1 from LADW.

3. Q2.8.3 Hint

3.1. Approach 1.

In solving the problem, we would like to apply the equation.

$$[I]_{SB}[I]_{BA} = [I]_{SA}$$

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where S is the standard basis $\{1, t\}$, A is the basis $\{1, 1+t\}$ and B is the basis $\{1-t, 2t\}$. Then we would have

$$A = [I]_{SA} = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$$

$$B = [I]_{SB} = \begin{bmatrix} 1 & 0 \\ -1 & 2 \end{bmatrix}$$

Notice that in writing vectors as column vectors etc, the order of the basis vectors is important. A change in the order of the basis would also affect the answer we get. Since the problem already gives an order for the basis B: $(1, 1+t)$, one should adhere to the order. If the student mistakenly exchanged the order of the basis vector here, then the student is likely to produce wrong answers.

Thus,

$$[I]_{BA} = B^{-1}A = \begin{bmatrix} 1 & 0 \\ \frac{1}{2} & \frac{1}{2} \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ \frac{1}{2} & 1 \end{bmatrix}.$$

These calculations are all made according to section 2.8.3 of LADW.

Any mistake is avoidable if the student checked their answer using the equation

$$[I]_{BA} = B^{-1}A$$

3.2. **Approach 2.** Even if we do not apply the above equation, notice that LADW describes the change of vector matrix as satisfying the following equation

$$[v]_B = [I]_{BA}[v]_A$$

We can solve the problem using only this equation.

Here $[v]_B$ and $[v]_A$ are the same vector, expressed under different bases, and thus as different column vectors. For example, if $v = 1$, then $[v]_A = [1, 0]^T$ and $[v]_B = [1, 1/2]^T$. Thus, the matrix $[I]_{BA}$ should be able to take the former column vector to the latter column vector. Thus, by selecting v such that $[v]_A = [1, 0]^T, [0, 1]^T$, and calculating the corresponding $[v]_B$, we are able to know what $[I]_{BA}$ is.

In some responses, the student interpret $[v]_B = [I]_{BA}[v]_A$ as if $[v]_A$ and $[v]_A$ are two vectors, expressed as the same column vector $[1, 0]^T$ or $[0, 1]^T$ under the two bases. For example, a student might write $[v]_A$ and $[v]_A$ as the second vector of A and B respectively. For example:

$$[v]_A = [1, 1]^T, [v]_B = [0, 2]^T$$

They would consequently yield wrong results.

Remember: (here is a slogan I just come up with) **vectors are absolute, while column vectors are relative** (to the basis selected). Column vectors are just representation of vectors under a specific basis selected. For example, if $v = 1$, then the representation of v under A and B, respectively, are $[v]_A = [1, 0]^T$ and $[v]_B = [1, 1/2]^T$.

The role of the change-of-coordinate matrix, then, is to transform the representation of a vector under one basis (A for example), to the representation of the same vector under another basis (B for example). This is the idea behind the equation

$$[v]_B = [I]_{BA}[v]_A.$$

4. Q2.8.4

When we change the basis, both input and output of the linear transformation T shall be expressed in terms of the new basis. Thus, the new matrix should capture the linear transformation from (a, b) to (c, d) in the following equation.

$$T(a[1,1]^T + b[1,2]^T) = c[1,1]^T + d[1,2]^T$$

In other words the matrix asked is

$$\begin{aligned}\begin{bmatrix}1 & 1\\1 & 2\end{bmatrix}^{-1}\begin{bmatrix}3 & 1\\1 & -2\end{bmatrix}\begin{bmatrix}1 & 1\\1 & 2\end{bmatrix} &= \begin{bmatrix}2 & -1\\-1 & 1\end{bmatrix}\begin{bmatrix}3 & 1\\1 & -2\end{bmatrix}\begin{bmatrix}1 & 1\\1 & 2\end{bmatrix}\\ &= \begin{bmatrix}5 & 4\\-2 & -3\end{bmatrix}\begin{bmatrix}1 & 1\\1 & 2\end{bmatrix}\\ &= \begin{bmatrix}9 & 13\\-5 & -8\end{bmatrix}\end{aligned}$$

5. Supplementary Problem 2

Consider the equation

$$\begin{bmatrix}a & b\\c & d\end{bmatrix}\begin{bmatrix}a & b\\c & d\end{bmatrix} = \begin{bmatrix}a^2+bc & b(a+d)\\c(a+d) & d^2+bc\end{bmatrix} = \begin{bmatrix}1 & 0\\0 & 1\end{bmatrix}$$

Because $a^2 = d^2$, we have either $a = d$ or $a = -d$.

When $a = d$, and a, d non-zero, we must have $b = c = 0$ and $a^2 = d^2 = 1$. There are 2 cases here.

When $a = -d$, there is no more restriction on b, c except for that

$$bc + a^2 = bc + d^2 = 1$$

For the first case stated above, there are possible matrices ($a = d = 1, 4$). For the second case, there are 30 possible matrices (4 for each cases when $a = 0, 2, 3$; 9 for each cases when $a = 1, 4$). Thus there are 32 possible matrices in total.